Credible Intervals

UBCO MDS — DATA 582

Review: Frequentist Confidence Intervals

- In the frequentist framework, the argument for confidence intervals starts with a probability statement, say

$$P(A < \bar{X} < B) = 0.95$$

- Since $\bar{X}$ is a random variable, this probability statement is fine. Knowing the sampling distribution of $\bar{X}$ allows us to replace A and B...

- So for large samples, we can say

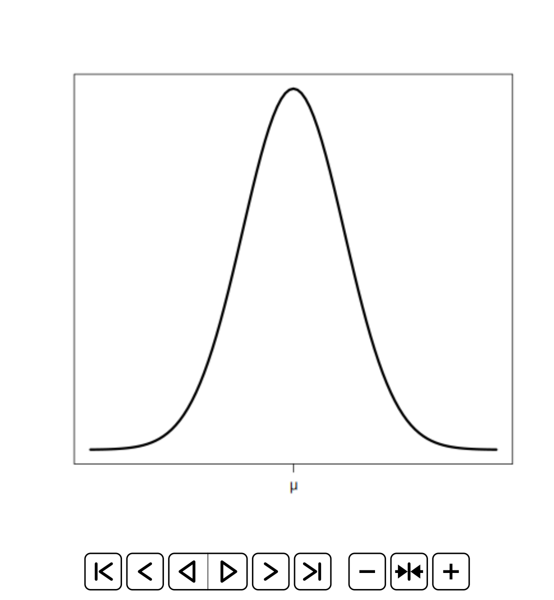
$$P\left(\bar{X} - 1.96\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 1.96\frac{\sigma}{\sqrt{n}}\right) = 0.95$$

- But why is it a problem to replace $\bar{X}$ with an observed $\bar{x}$?

Review: Frequentist Confidence Intervals

- This leads us to be very careful with the interpretation of a computed confidence interval!!
- "We are 95% confident that our computed confidence interval $\bar{x} \pm 1.96\frac{\sigma}{\sqrt{n}}$ contains the true parameter μ."
- This phrasing is a clever-ish way to not make an explicit probability statement. A more wordy and accurate interpretation is...
- "If we sampled n observations repeatedly from the same population (X), and computed the interval each time in the same fashion, 95% of those intervals would contain the true parameter μ."

Frequentist Confidence Intervals: Animated



Review: Frequentist Confidence Intervals

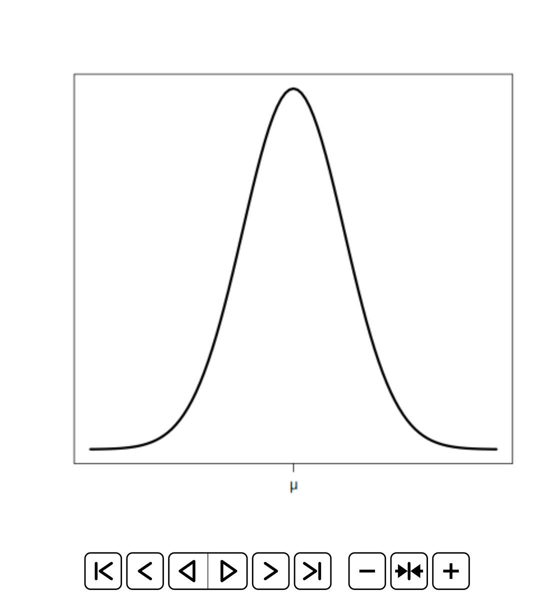
- While you might be used to equal tail intervals like we just reviewed, nothing keeps us from changing A and B in the probability statement.

$$P(A < \bar{X} < B) = 0.95$$

- Suppose I want a CI that favours guessing above $\bar{X}$ for some reason...

$$P\left(\bar{X} - 1.65\frac{\sigma}{\sqrt{n}} < \mu < \bar{X} + 3\frac{\sigma}{\sqrt{n}}\right) = 0.95$$

Frequentist Confidence Intervals: Animated



Review: Frequentist Confidence Intervals

- So, if this is also just as relevant a confidence interval from a frequentist standpoint, why have most of you probably never computed one like this?
- What's the benefit of the equal tail interval versus a lopsided one (at least in the normal mean case)?

Bayesian Credible Intervals

- In the Bayesian paradigm, inferential intervals are called credible intervals
- This is to largely distinguish them from frequentist confidence intervals, as while their goals are similar the end results are a bit different.
- They can generally be built in a straightforward manner from the posterior distribution

Bayesian Credible Intervals

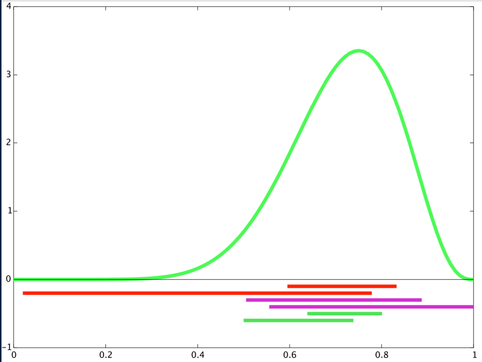
- A Bayesian credible interval, as in the frequentist paradigm, starts with a probability statement. But this time, since our parameter is a RV, we can jump straight to

$$P(A < \theta < B) = 0.95$$

- Notably, if we replace A and B with appropriate values from $h(\theta \mid \mathbf{x})$ that satisfy the statement, the probabilistic statement still applies!
- Because θ is a RV, the 95% credible interval (A, B) contains θ with probability 0.95.
- This is a massive simplification for our interpretation.

Credible Interval Width

- Similar to my lopsided example in the frequentist world, we have infinitely many ways to define a credible interval if $h(\theta \mid \mathbf{x})$ is continuous.



- All the above are 68% credible intervals (Source for image)

Credible Interval Typical Choices

- The simplest, and most common, is to use equal-tail quantiles just as you're used to in the frequentist case
- So for a 95% (or 0.95 probability) credible interval, we split 0.05 between the two tails. The typical notation from intro stats applied to this would be

$$P(\theta_{0.025} < \theta < \theta_{0.975}) = 0.95$$

- Supposing we were doing a 95% credible interval for μ and the posterior happened to be standard normal then

$$P(-1.96 < \mu < 1.96) = 0.95$$

Credible Interval Typical Choices

- For skewed/truncated and other messy-form distributions, the equal-tail approach may lead to wider intervals than other choices that provide the same probability.
- We can instead pick A and B from $P(A < \theta < B) = 0.95$ such that we minimize $B - A$ (aka, the width of the interval).
- This minimum width interval can be found by focusing on high probability density for the posterior. It is therefore called the high posterior density, or HPD, interval.
- These are unique, and contain the mode, if the posterior is unimodal. Finding them is a (somewhat easy) optimization problem.

Example: Return to Bayes-ball

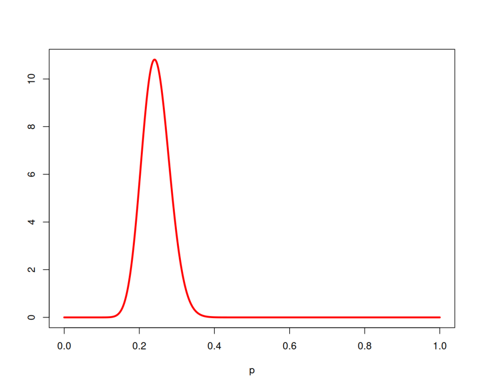
- Recall our framework for our MLB rookie Jacob...
  - At bats: $X \sim \text{Bernoulli}(p)$
  - Prior: $p \sim \text{Beta}(30, 100)$
  - Posterior $p \mid \mathbf{x} \sim \text{Beta}(30 + \sum x_i, 100 + n - \sum x_i)$
- Again, we observe his debut MLB game, and he gets 3 hits on 5 at bats...

Example: Return to Bayes-ball

- In the frequentist framework, we knew $\hat{p} = \frac{3}{5} = 0.6$ as the typical estimator
- What about a 95% confidence interval?
- $n = 5$ means that CLT does not apply. Uh oh...
- Well, there are still options, but it's not worth us covering the details. Most approaches would provide us something like (0.2, 0.9) as a 95% CI

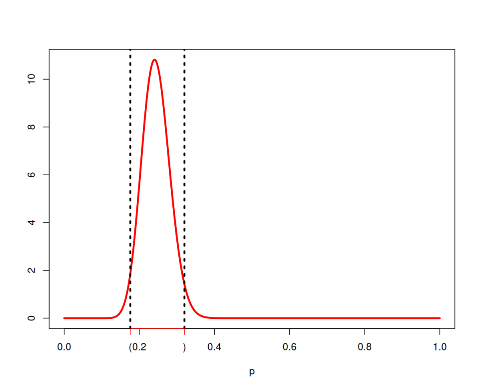
Example: Return to Bayes-ball

- In the Bayesian framework, we find the posterior to be Beta$(33, 102)$



Example: Return to Bayes-ball

- We find, through software, $p_{.025} = 0.176$ and $p_{.975} = 0.320$.



Example: Return to Bayes-ball

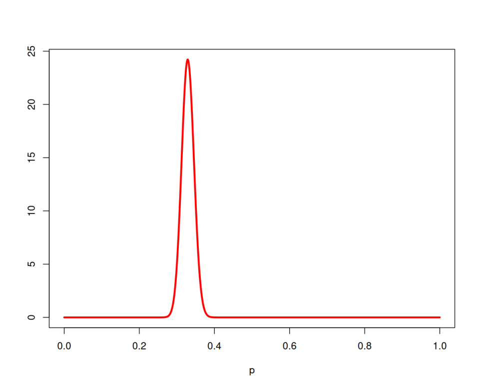
- Frequentist: "We are 95% confident that Jacob's true batting average is somewhere between (0.2, 0.9)."
- Bayesian: "There is a 95% chance that Jacob's true batting average falls somewhere between (0.176, 0.320)."
- Obviously, the Bayesian interval is much more narrow, as it is relying on strong prior information about what we expect batting averages to look like.
- But focus on the wording. 95% confident is not a direct probability statement, whereas 95% chance is.

Example: Return to Bayes-ball

- For Jacob's first season, in the frequentist framework, we find $\hat{p} = \frac{238}{684} = 0.348$ as the typical estimate
- What about a 95% confidence interval?
- $n = 684$ means that CLT does apply.
- So the typical CI approach leads us to (.307, .378)

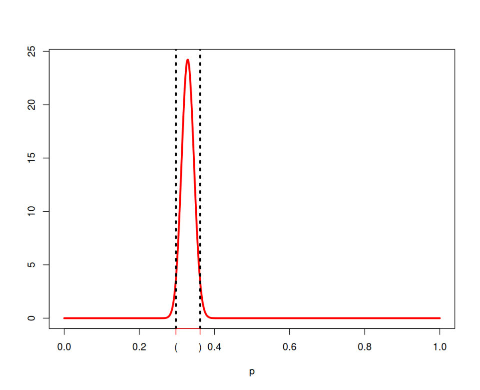
Example: Return to Bayes-ball

- In the Bayesian framework, we find the posterior to be $\text{Beta}(268, 546)$



Example: Return to Bayes-ball

- We find, through software, $p_{.025} = 0.297$ and $p_{.975} = 0.362$.



Example: Return to Bayes-ball

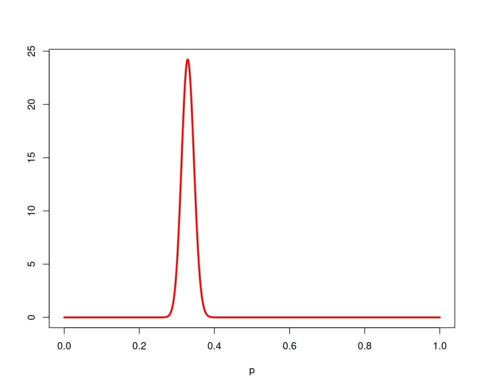
- So after the first season...
- Frequentist: "We are 95% confident that Jacob's true batting average is somewhere between (.307, .378)."
- Bayesian: "There is a 95% chance that Jacob's true batting average falls somewhere between (0.297, 0.362)."
- Now, the intervals have gotten much more similar in width and location.

Example: Caution

- What if our prior was strong, but poorly specified
  - At bats: $X \sim \text{Bernoulli}(p)$
  - Prior: $p \sim \text{Beta}(100, 30)$
  - Posterior $p \mid \mathbf{x} \sim \text{Beta}(100 + \sum x_i, 30 + n - \sum x_i)$
- We'll jump straight away to the season with 238 hits at 684 at bats.

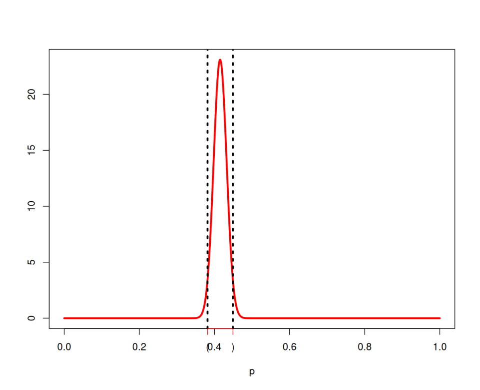
Example: Caution

- In the Bayesian framework, we find the posterior to be Beta$(338, 476)$



Example: Caution

- We find, through software, $p_{.025} = 0.382$ and $p_{.975} = 0.450$.



Example: Caution

- So after the first season...
- Frequentist: "We are 95% confident that Jacob's true batting average is somewhere between (.307, .378)."
- Bayesian: "There is a 95% chance that Jacob's true batting average falls somewhere between (0.382, 0.450)."
- But note that Jacob's season-long observed average is 0.342, which doesn't even fall into the Bayesian credible interval!!
- So clearly, care needs to be taken when specifying strong priors.

Inference about μ

- Let's change gears to another common Bayesian inference framework
- Suppose $X \sim N(\mu, \sigma^2)$, and we assume σ^2 is known.
- We target the population mean μ for inference, and place a prior on it to also be normal: $\mu \sim N(M, S^2)$
- Let's seek the posterior distribution for μ.

Inference about μ

- We need $h(\mu \mid \mathbf{x}) \propto f(\mathbf{x} \mid \mu) g(\mu)$

$$
\begin{aligned}
f(\mathbf{x} \mid \mu) &= \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left\{-\frac{1}{2\sigma^2}(x_i - \mu)^2\right\} \\
&= (2\pi\sigma^2)^{n/2} \exp\left\{-\frac{1}{2}\sum_{i=1}^{n}\left(\frac{x_i - \mu}{\sigma^2}\right)^2\right\} \\
&\propto \exp\left\{-\frac{1}{2}\sum_{i=1}^{n}\left(\frac{x_i - \mu}{\sigma^2}\right)^2\right\} \\
&\propto \exp\left\{-\frac{1}{2}\frac{(\mu - \bar{x})^2}{\sigma^2/n}\right\}
\end{aligned}
$$

Inference about μ

- We need $h(\mu \mid \mathbf{x}) \propto f(\mathbf{x} \mid \mu) g(\mu)$

$$
\begin{aligned}
g(\mu) &= \frac{1}{\sqrt{2\pi S^2}} \exp\left\{-\frac{1}{2S}(\mu - M)^2\right\} \\
&\propto \exp\left\{-\frac{1}{2S}(\mu - M)^2\right\}
\end{aligned}
$$

Inference about μ

- We need $h(\mu \mid \mathbf{x}) \propto f(\mathbf{x} \mid \mu) g(\mu)$

$$h(\mu \mid \mathbf{x}) \propto \exp\left\{-\frac{1}{2}\frac{(\mu - \bar{x})^2}{\sigma^2/n}\right\} \exp\left\{-\frac{1}{2S}(\mu - M)^2\right\}$$

$\vdots$ details provided in walkthrough

$$\propto \exp\left\{-\frac{(\mu - b/a)^2}{2(1/a)}\right\}$$

- where $a = \frac{1}{S^2} + \frac{n}{\sigma^2}$ and $b = \frac{M}{S^2} + \frac{\sum x_i}{\sigma^2}$.

Inference about μ

- We need $h(\mu \mid \mathbf{x}) \propto f(\mathbf{x} \mid \mu) g(\mu)$

$$h(\mu \mid \mathbf{x}) \propto \exp\left\{-\frac{1}{2}\frac{(\mu-\bar{x})^2}{\sigma^2/n}\right\} \exp\left\{-\frac{1}{2S}(\mu - M)^2\right\}$$

$$\vdots \quad \text{details provided in walkthrough}$$

$$\propto \exp\left\{-\frac{(\mu - b/a)^2}{2(1/a)}\right\}$$

- where $a = \frac{1}{S^2} + \frac{n}{\sigma^2}$ and $b = \frac{M}{S^2} + \frac{\sum x_i}{\sigma^2}$.

Inference about μ

- Now, same as in the Beta-Bernoulli framework, we need to seek the functional form of this

$$h(\mu \mid \mathbf{x}) \propto \exp\left\{-\frac{(\mu - b/a)^2}{2(1/a)}\right\}$$

- where $a = \frac{1}{S^2} + \frac{n}{\sigma^2}$ and $b = \frac{M}{S^2} + \frac{\sum x_i}{\sigma^2}$.

- With a bit of thinking, we can see where μ here is the random variable, it follows the functional form of $N(\frac{b}{a}, \frac{1}{a})$

- Thus, we now have the Bayesian framework for the first inferences most of you would have learned in the frequentist paradigm: $X \sim N(\mu, \sigma^2)$ with σ^2 known!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